



← All roadmaps

MLOPS MENA COMMUNITY

Basic MLOps Engineer Roadmap

From zero to job-ready

Beginner → Job-ready

🕒 6-9 months

📅 10-15 hrs/week

📖 51 free resources

👤 Students, ML Engineers, Software Engineers, and Data Scientists breaking into MLOps.

[Read the original on LinkedIn ↗](#)



PHASE 0
Month 1-2



PHASE 1
Month 3



PHASE 2
Month 4-5



PHASE 3
Month 6

This is the roadmap I'd give my younger self — focused on free, open-source resources because the real knowledge in this field lives on GitHub and YouTube, not behind expensive paywalls.

If you're a student, ML Engineer, Software Engineer, or Data Scientist looking to break into MLOps, this is for you. Realistic timeline: 6-9 months at 10-15 hours/week.



Phase 0 — Foundations (Month 1-2)

Before you touch any MLOps tool, build a solid foundation in four areas.

Python

OOP, decorators, virtual environments, type hints, numpy/pandas/scikit-learn.

- [freeCodeCamp Python Course \(YouTube\)](#)
- [Real Python](#)

Linux & Bash

Every production server runs Linux. You need to be comfortable on the command line.

- [MIT "The Missing Semester" \(official site\)](#)
- [Missing Semester YouTube playlist](#)

Git & GitHub

Branching, merging, pull requests, working in a team.

- ["Pro Git" book \(free\)](#)
- [freeCodeCamp Git & GitHub tutorial](#)

ML fundamentals

You don't need a PhD, but you need to understand the basics: supervised vs unsupervised, train/val/test splits, overfitting, evaluation metrics.

- [Andrew Ng's "Machine Learning Specialization" \(Coursera, audit free\)](#)
- [StatQuest with Josh Starmer \(YouTube\)](#)



Phase 1 — Software Engineering for ML (Month 3)

This is what separates an MLOps Engineer from a Jupyter notebook user.

Docker

Containerizing applications is the #1 MLOps skill.

- [TechWorld with Nana \(YouTube channel\)](#)
- [Docker crash course \(3 hours\)](#)
- [Docker official docs](#)

FastAPI

Wrapping your model as a REST API. FastAPI's official tutorial is the clearest documentation I've ever seen — pair it with ArjanCodes on YouTube.

- [FastAPI official tutorial](#)
- [ArjanCodes \(YouTube\)](#)

Testing with pytest

Writing unit and integration tests for ML code.

- [pytest official docs](#)
- ["Python Testing with pytest" book](#)



Take your Phase 0 model, wrap it in FastAPI, containerize it with Docker, write tests, and push to GitHub.

Phase 2 — MLOps Core (Month 4-5)

If you take only ONE thing from this entire roadmap, take this:  **MLOps Zoomcamp by DataTalks.Club** — a completely free course on GitHub covering experiment tracking, orchestration, deployment, and monitoring, with hands-on projects and an active Slack community.

- [MLOps Zoomcamp GitHub](#)
- [MLOps Zoomcamp YouTube playlist](#)
- [DataTalks.Club Slack](#)

Within Phase 2, you'll cover the following.

Data versioning — DVC

Git for large datasets and models. Start with DVC's official YouTube channel and the interactive tutorials on dvc.org.

- [DVC official site](#)
- [DVC YouTube channel](#)

Experiment tracking — MLflow

Track every experiment, compare runs, manage model versions.



Orchestration — Apache Airflow

Schedule and orchestrate your data and ML pipelines. Marc Lamberti is the Airflow guru — his free crash course (4 hours) covers everything you need, and Astronomer's Airflow Academy is free with certification.

- [Marc Lamberti's YouTube \(Data with Marc\)](#)
- [Airflow official docs](#)
- [Astronomer Academy \(free Airflow certification\)](#)

Model serving — FastAPI + BentoML

Build production-ready inference APIs.

- [BentoML docs](#)
- [BentoML GitHub](#)

CI/CD — GitHub Actions

Automate your testing, building, and deployment pipelines.

- [GitHub Actions docs](#)
- [GitHub Actions YouTube tutorial \(TechWorld with Nana\)](#)

Bonus open-source gem

"Made With ML" by Goku Mohandas — 35K+ stars, a complete MLOps course with code, free.

- ["Made With ML" \(GitHub\)](#)
 - [Made With ML website](#)
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Phase 3 — Cloud Basics (Month 6)

Pick ONE cloud provider in the beginning. Don't try to learn three at once.

AWS — most common in the MENA region (especially KSA & UAE)

- freeCodeCamp's AWS Cloud Practitioner (14 hours)
- AWS Skill Builder (free tier)

GCP — best ML services (Vertex AI, BigQuery ML)

- Google Cloud Skills Boost (free labs after sign-up)

Azure — common in enterprise companies

- Microsoft Learn

Focus on object storage (S3/GCS), compute (EC2/Compute Engine), container services, and ML-specific services (SageMaker/Vertex AI).

Phase 4 — Basic Monitoring (Month 7)

Evidently AI

Open-source toolkit for data drift and model monitoring. Their YouTube channel has full workshops.

- Evidently AI site
- Evidently AI YouTube
- Evidently AI GitHub



The standard for infrastructure monitoring. TechWorld with Nana has great tutorials here too.

- [Prometheus docs](#)
- [Grafana docs](#)

YouTube Channels You MUST Subscribe To

MLOps & ML engineering

- [DataTalks.Club](#)
- [MLOps Community](#)
- [DeepLearning.AI](#)

DevOps tools

- [TechWorld with Nana](#)
- [KodeKloud](#)

ML intuition

- [StatQuest with Josh Starmer](#)
- [3Blue1Brown](#)

Free Newsletters & Blogs

- [The Batch \(Andrew Ng\)](#)



- Sebastian Raschka's "Ahead of AI"

My Honest Advice From Personal Experience

- **Build projects, don't just watch videos.** End every phase with a GitHub project. Your portfolio matters more than any certificate.
- **Contribute to open source.** One accepted PR to MLflow or Evidently is a strong CV point.
- **Join communities.** MLOps Community Slack, Hugging Face Discord, and of course the MLOps MENA Community we're building together.
- **Write about what you learn.** LinkedIn or Medium. Writing crystallizes knowledge and builds your personal brand.
- **Don't rush the foundations.** Tools change, concepts last.

Other paths



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Mid → Sr

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The DevOps to MLOps Transition Roadmap

The fastest career pivot in tech

5 phases · 3–5 months

MLOps Roadmap for Seniors

Depth, system thinking, and production
scale

7 phases · Ongoing

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Brainsmingle space

Free to join this week with code **FREECORETEAM**

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